



Multitask Learning for Point-of-Interest Classification and Prediction Tasks: The Role of the Check-in-Level Representation

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Sumário

- ❶ **Abertura — a pergunta e o escopo**
- ❷ Fundamentos compartilhados
- ❸ Multitask Learning for POI Category and Next-POI Prediction
- ❹ ST-MTLNet: Spatio-Temporal POI Representations
- ❺ **A Check-in-Level Multitask Study of Next Category and Region**
- ❻ Conclusão Geral — a resposta condicional

Human Mobility

- ▶ Human movement is highly regular;
- ▶ people return to a small set of places, and travel far less often;
- ▶ potential predictability of an individual's next location at about **93 percent** (Song et al., 2010).

The caveat, in the same breath

That estimate is for the next **location** at a coarse spatial resolution. It shows that mobility contains learnable regularity, and it is **not a ceiling** for the metrics reported in this work.

- ▶ What a service acts on, once the next visit is anticipated: recommendation, navigation, transit planning, allocation by area.

The ground: check-ins and what joint training promises

- ▶ **LBSN** (Silva et al., 2019) — a location-based social network; its record is the **check-in**: a user, a place (point of interest, POI), and a time;
- ▶ **multitask learning (MTL)** (Caruana, 1997) — train related tasks together so they can share information;
- ▶ the operational appeal here: **one model to maintain, one forward pass, both predictions.**

No guarantee

Joint training does not guarantee better predictions. Shared parameters can harm one task, a failure known as **negative transfer**.

The question

Does multitask learning help next-category and next-region prediction, and what does the answer depend on?

Three studies, in sequence

- ▶ **Ch. 3 — Multitask Learning for POI Category and Next-POI Prediction.** CBIC 2025, DOI 10.21528/CBIC2025-1191324. First author;
- ▶ **Ch. 4 — ST-MTLNet: Spatio-Temporal POI Representations.** CoUrb 2026 (SBRC workshop), DOI 10.5753/courb.2026.22960. Contributed the MTLnet baseline; presented at the event;
- ▶ **Ch. 5 — A Check-in-Level Multitask Study of Next Category and Region.** MobiWac 2026, accepted. First author.

“The first reports a negative result, the second identifies its main bottleneck, and the third tests the resulting solution.”

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The tasks

	reads	predicts
Category classification (2.6)	one place	that place's category
Next category (2.7)	a history of N visits	the category of the next visit
Next region (2.8)	a history of N visits	the neighborhood-scale unit
Next place (2.9)	—	<i>not predicted in this work</i>

Naming trap

In Chapters 3 and 4, “**Next-POI Prediction**” means the **next category**, not the exact next place.

The pair changes: Ch. 3–4 category classification + next category; Ch. 5 next category + next region.

Related work: POI prediction

- ▶ **POI** — a place, with a category and a region;
- ▶ **Check-in** — one recorded visit: a user, a place, a time.
- ▶ **Next place** — the literature also calls it **next-POI**, **next-location**, and they are the same task. It is the **dominant target in the field** — recurrent: ST-RNN (Liu et al., 2016), DeepMove (Feng et al., 2018), Flashback (Yang et al., 2020); attention: STAN (Luo et al., 2021), GeoSAN (Lian et al., 2020), GETNext (Yang et al., 2022);
- ▶ every model named there predicts the **exact establishment**, so **none is a direct baseline** for the targets studied here.

Category and region: means or end

Category and region as a MEANS

Toward the next place: HMT-GRN (Lim et al., 2022), CatDM (Yu et al., 2020), CSLSL (Huang et al., 2024).

Category or region as an END

Activity region (DRRGNN, Zhu et al., 2022), next category (POI-RGNN, Capanema et al., 2023).

Among the works reviewed in this dissertation, none treats the next category and the next region as co-equal end targets of one joint model that does not also predict the next place.

next category	POI-RGNN · Markov (Gambs et al., 2012) over category transitions
next region	HMT-GRN · STAN · ReHDM (Li et al., 2025) · Markov-1 floor
Ch. 3 · Ch. 4	HMRM (Chen et al., 2020) · MHA+PE (Zeng et al., 2019) · MTLnet

MTL Fundamentals

- ▶ **Hard parameter sharing** (Def. 2.10) — one shared trunk; tasks separate only at their own output.
Soft parameter sharing (Def. 2.11) — one network per task, coupled by a penalty on their differences;
- ▶ **Negative transfer** (Def. 2.12): joint training leaves a task worse than its dedicated single-task model;
- ▶ **Balancing methods** — two classes:
set the weights — uncertainty weighting, GradNorm, DWA, FAMO;
change the update direction — MGDA, PCGrad, CAGrad, Nash-MTL, Aligned-MTL;
- ▶ **Pareto dominance** — no worse on every task loss, better on at least one;
- ▶ **MTL in mobility: next place, almost entirely** — MCARNN, iMTL, HAMTL.

This dissertation claims no Pareto property for its models.

How places are represented

- ▶ **one-hot identifier** — marks a place or a category by position, and encodes no relationship between identifiers;
- ▶ **skip-gram** (Mikolov et al., 2013), **DeepWalk** (Perozzi et al., 2014), **node2vec** (Grover & Leskovec, 2016) — dense vectors whose geometry reflects relationships in the data;
- ▶ **GCN** (Kipf & Welling, 2017), **GAT** (Veličković et al., 2018), **GraphSAGE** (Hamilton et al., 2017) — those relationships learned through neighborhood aggregation;
- ▶ **contrastive infomax** — an approach that builds representations from two pairs, one positive and one corrupted, so no labels are needed to train.

The substrate the three have in common

In all three, a **Delaunay triangulation** joins the places. The **edge weights differ**, and each chapter states its own.

The evidence base: six datasets

Dataset	Source	Check-ins	Regions	Majority (%)	Used by
Istanbul	Massive-STEPS (Wongso et al., 2025)	462,615	520	33.4	Ch. 5
AL	Gowalla (Cho et al., 2011)	113,846	1,109	34.2	Ch. 5
AZ	Gowalla	236,450	1,547	34.0	Ch. 5
FL	Gowalla	1,407,034	4,703	24.7	Ch. 5
TX	Gowalla	4,089,892	6,553	31.0	Ch. 4, Ch. 5
CA	Gowalla	3,171,380	8,501	32.7	Ch. 4, Ch. 5

The seven categories, identical in all three studies: Community, Entertainment, Food, Nightlife, Outdoors, Shopping, Travel.

Region — a census tract in the five U.S. datasets, a *mahalle* in Istanbul; both partition a city at neighborhood scale.

Majority — share of next-visit labels in the most common category, which is **Food** in every dataset.

The metric all three studies share

- ▶ **macro-F1** — the mean of the per-category F1 scores; every category counts equally, so rare ones matter. Out of 100;
 - ▶ **why** — the class distribution is imbalanced (**Food is roughly one third** of the check-ins in a representative state), so plain accuracy hides poor performance on rare classes;
 - ▶ **what it does not do** — it does not show *which* classes improve, and it can be low while overall accuracy is high.

Three reference points, not competitors

Majority class — a predictor that always answers the most common category.

Dedicated model — the task's own head, trained alone on the same input, without the shared trunk.

Markov — a transition table over the training visits: it answers with what most often follows what. **First order for region; strongest order per dataset for category.**

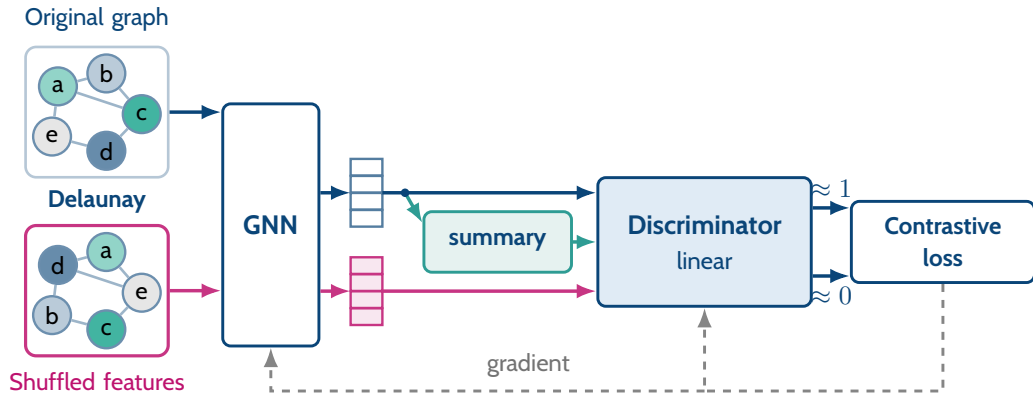
The protocol of the first two studies

- ▶ **Ch. 3 and 4:** five-fold CV stratified **over samples** — **one user's check-ins may fall on both sides**;
- ▶ **non-overlapping windows of N visits**; users with fewer than five visits are discarded; **one seed**;
- ▶ Full epoch budget, no early stopping; each task at **its own best epoch**;
- ▶ Fold means and standard deviations, **no significance tests**;
- ▶ *outperforms* is reserved for a paired superiority test, and these two chapters report none — **so these two chapters report differences, never a verdict.**

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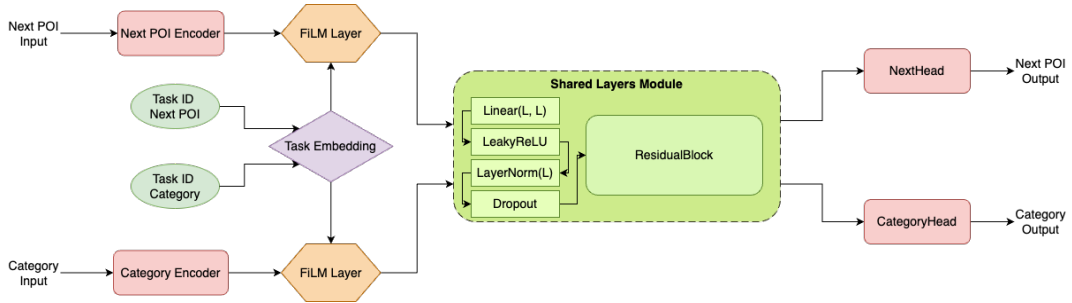
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Deep Graph Infomax (DGI)



- **Node features, as released** (Veličković et al., 2019) — the mean of the one-hot vectors of a place's graph neighbors, with the place's own vector **excluded**;
- **What it gives** — one **64-dimensional vector per place**: every visit enters with the same vector.

MTLnet



- ▶ **FiLM** (Perez et al., 2018): a learnable task embedding gives a **scale** and a **shift**: both tasks read the same parameters under different scalings.
- ▶ **Shared residual blocks** = **hard parameter sharing** (Def. 2.10). Two disjoint parameter sets: **shared** · **task-specific**.
- ▶ **Heads** — category: **three parallel MLPs**; next category: **Transformer encoder**.

Nash-MTL

- ▶ Changes the **update direction** — gradient combination as a **cooperative bargaining problem**;
- ▶ Each task's utility is its own loss reduction; the direction **maximizes the product**;
- ▶ Guarantee: a **Pareto-stationary point** — necessary, not sufficient.

What Chapter 3 claims. Lower combined multitask loss than PCGrad and than no such optimizer; the chapter adopted it. **A conclusion of the time**, weakened by a later finding.

The null result

Per-category F1; $\pm$ is the standard deviation over the five folds.

Category classification (Def. 2.6)

	MTL	Single	HMRM
Community	<u>52.13 $\pm$ 0.90</u>	53.11 $\pm$ 0.58	20.25 $\pm$ 0.52
Entertainment	<u>41.44 $\pm$ 1.29</u>	41.73 $\pm$ 1.65	13.07 $\pm$ 1.14
Food	57.43 $\pm$ 1.46	<u>56.70 $\pm$ 0.84</u>	28.44 $\pm$ 0.42
Nightlife	<u>29.94 $\pm$ 2.27</u>	34.22 $\pm$ 2.08	25.54 $\pm$ 1.73
Outdoors	<u>47.19 $\pm$ 1.01</u>	47.75 $\pm$ 0.89	16.11 $\pm$ 0.68
Shopping	62.51 $\pm$ 0.94	<u>61.88 $\pm$ 1.01</u>	46.69 $\pm$ 0.81
Travel	<u>43.16 $\pm$ 1.43</u>	43.20 $\pm$ 1.94	15.25 $\pm$ 1.19

Next category — the leads split

	MTL	Single	MHA+PE
Community	34.79 $\pm$ 1.05	<u>34.90 $\pm$ 0.61</u>	35.83 $\pm$ 1.70
Entertainment	24.21 $\pm$ 1.75	26.06 $\pm$ 1.01	<u>25.48 $\pm$ 0.86</u>
Food	<u>28.06 $\pm$ 3.55</u>	26.35 $\pm$ 2.97	43.47 $\pm$ 0.50
Nightlife	22.07 $\pm$ 0.52	<u>21.79 $\pm$ 0.30</u>	1.55 $\pm$ 1.50
Outdoors	<u>21.61 $\pm$ 0.50</u>	22.45 $\pm$ 0.81	12.56 $\pm$ 1.94
Shopping	<u>42.58 $\pm$ 2.06</u>	42.18 $\pm$ 1.13	43.53 $\pm$ 1.66
Travel	64.61 $\pm$ 1.11	<u>64.32 $\pm$ 0.88</u>	26.91 $\pm$ 0.97

"largely comparable, without a clear or consistent advantage for the multitask learning setup in these experiments"

The null result: three possible causes

1. Task dissimilarity

A static task and a sequential task may force the shared trunk into a **compromise representation**, specialized for neither.

2. Representation insufficiency

The shared representation may not be rich enough to encode **semantic properties and sequential dynamics at once**.

3. Topology rigidity

One shared block may be too restrictive; **soft sharing or Mixture-of-Experts models** might fit better.

Negative transfer was hypothesized here. It was never observed.

Three possible causes do not close the investigation.
It designs the next experiment:

keep the architecture fixed, and move only the input.

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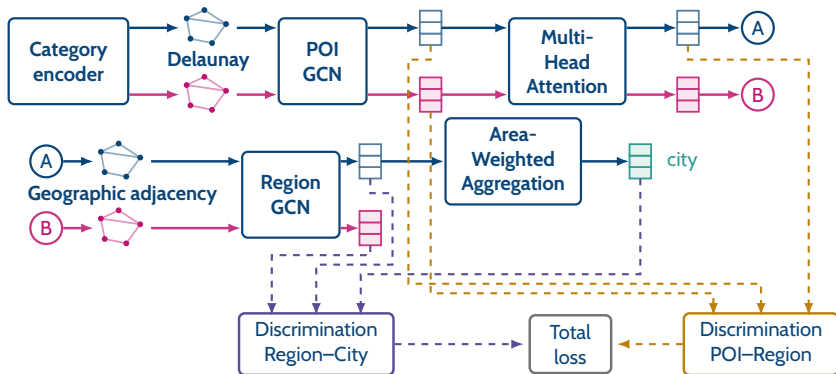
Architecture or representation?

The inherited question

Is the bottleneck the **representation**, or the **sharing topology**?

- ▶ **The design that separates them** — MTLnet unchanged, **only the input moves**: same trunk, same FiLM, same balancer, same hyperparameters;
- ▶ **old input** — one monolithic **64-dimensional** place embedding (DGI);
- ▶ **new input** — three **64-dimensional** encoders, spatial, temporal and categorical, concatenated into **192**;
- ▶ **Both models share the same 256-dimensional latent width**: each task encoder projects whatever it receives to 256, so the trunk and the heads are identical. **What differs is the input: 192 dimensions against 64.**

Hierarchical Graph Infomax (HGI)



- Repurposed here — a use the original evaluation does not cover.
- Preprocessing — places → **Delaunay** graph → **Node2Vec** walks (Grover & Leskovec, 2016) → **skip-gram** with negative sampling (Mikolov et al., 2013) + a **hierarchical category loss** (Xu et al., 2023) → one **64-d** vector per place type → **HGI** (Huang et al., 2023).

Why these encoders

Spatial, 64 d

Why: space entered only through graph topology, never as a coordinate.

- ▶ **SIREN** (Sitzmann et al., 2020) — sinusoidal activations over normalized coordinates;
- ▶ **Sphere2Vec-M** (Mai et al., 2023) — multi-scale, spherical;
- ▶ one contrastive loss for both, on geographic distance.

Temporal, 64 d

Why: no explicit temporal representation.

Time2Vec (Kazemi et al., 2019) — a linear term for trend, sinusoidal terms for hour of day and day of week.

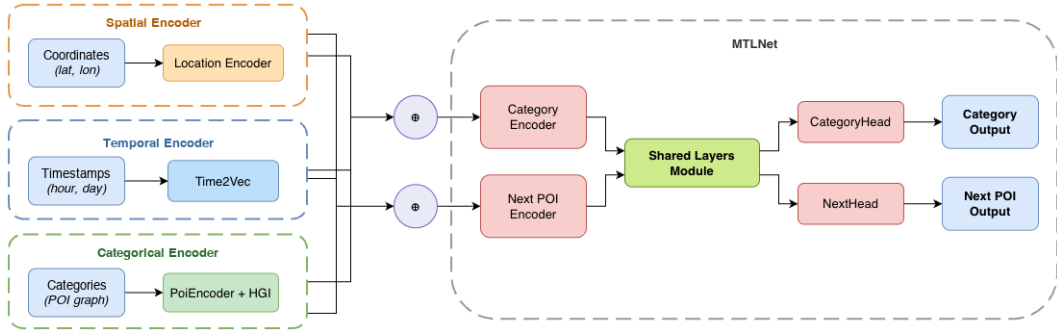
Categorical, 64 d, two phases

Why: graph structure alone — no hierarchy, no region.

1. **POI Encoder** — skip-gram with negative sampling.
2. **HGI** — the regional hierarchy above.

Architecture or representation?

The same MTLnet, with the decomposed input in place of the monolithic one



► **Heads unchanged from MTLnet** — only the input moves.

The diagnostic result is the sequential task

Florida — both tasks, same models

Category classification (Table 6)

	MTLnet	SIREN	Sph2Vec-M
Community	51.86	70.00	<u>69.84</u>
Entertainment	41.24	64.45	<u>63.92</u>
Food	55.47	<u>72.92</u>	73.22
Nightlife	32.59	62.60	<u>62.44</u>
Outdoors	47.71	<u>65.64</u>	66.35
Shopping	62.96	77.48	<u>77.47</u>
Travel	45.49	64.89	<u>64.77</u>

Next category (Table 7)

	MTLnet	SIREN	Sph2Vec-M
Community	34.33	38.31	<u>37.98</u>
Entertainment	25.34	31.38	<u>31.24</u>
Food	27.12	41.91	<u>41.65</u>
Nightlife	21.33	<u>23.32</u>	23.98
Outdoors	21.61	21.29	<u>21.59</u>
Shopping	39.62	<u>44.00</u>	44.66
Travel	64.47	<u>45.00</u>	44.93

Observation — the decomposed input holds the higher mean in **15 of 21** combinations across the three states; **MTLnet** retains six.

Interpretation — gains concentrate at **Food**, in all three states, and reach +20.2 to +22.0 pp per state, taking the better encoder in each cell.

The static task's input contains the label it predicts — the venue-type feature maps one-to-one onto the seven categories — so the left-hand numbers measure a **lookup**, not learned semantic inference. The sequential task, on the right, is the diagnostic result.

Three limits of the decomposition

What moved, and what did not

The decomposition helps say what a place is. On *Travel*, it does not help say where the user **goes** next: 45.49 → **64.89** on category classification at Florida, while on next category MTLnet keeps **64.47** against 45.00.

No universally better spatial encoder

SIREN at Florida and California,
Sphere2Vec-M at Texas.

Not width-matched

192 dimensions against 64. The chapter asks for an equalised-dimension control, and **it was never run.**

The three components are used together; this chapter does not isolate each encoder.

Freeze the architecture, move only the input, and the result moves.

The representation is the bottleneck.

Two limits stay: one vector per **place**, and a split that shares users.

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Three changes, each a consequence of the diagnosis

Not a preference — each follows from the previous chapter

	From → To	Why
Representation	place level → check-in level	weekday lunch $\neq$ Saturday night
Topology	hard parameter sharing → cross-attention	private spatial path for region
Protocol	sample-stratified → user-disjoint	four seeds, tests fixed in advance

The task pair changes here

Under a check-in-level representation, a second sequential target is a more natural companion than a static one: the pair becomes **next category** + **next region**.

The article this chapter reproduces was **accepted at MobiWac 2026** after the dissertation was deposited; the deposited volume records it as under review.

Next region: the task, and why it is worth predicting

Related work of this study, part 1 of 2

- ▶ **The target** — a region covers a larger area than a place. It is an easier target than the exact place, but the task is not easy;
- ▶ **Why this target** — the category says *what type* comes next; the region says *where*, hence where to prepare;
- ▶ **Where it sits** — the standard formulation targets a **grid cell**; here, official neighborhood-scale units. In MCMG (Sun et al., 2024) and HMT-GRN, category and region are **auxiliary** to a primary next-place task;
- ▶ **Practical applications** — demand and load anticipation, caching ahead of time, capacity planning. **No such service is built or evaluated here.**

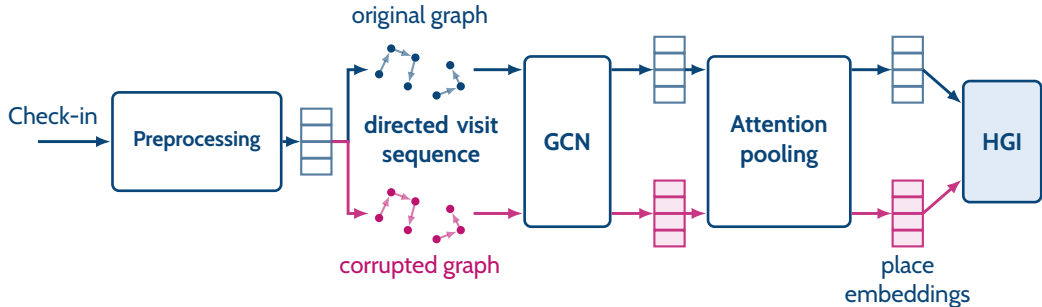
Why a per-visit representation is new in this line

Related work of this study, part 2 of 2

- ▶ **CTLE** (Lin et al., 2021) — the closest prior contextual check-in representation: one vector per visit, learned by masking and reconstructing a user's check-in sequence, from **place identifiers and timestamps alone**;
- ▶ **the difference is the construction** — CTLE is a **sequence model**; Check2HGI stays a **graph model**: same hierarchy, same infomax objective, **one level deeper**.

The novelty is the combination: per-visit context **inside** a hierarchical graph-infomax representation.

Check2HGI: a fourth level below the place



- **Preprocessing** — the check-in itself → category indicator + **sine and cosine** of hour of day and day of week + four **elapsed-time** values = **15 values per visit** → forward-only visit graph → **Check2HGI** exports one **64-d** vector per visit.

What each visit contributes

The three groups, and one design principle

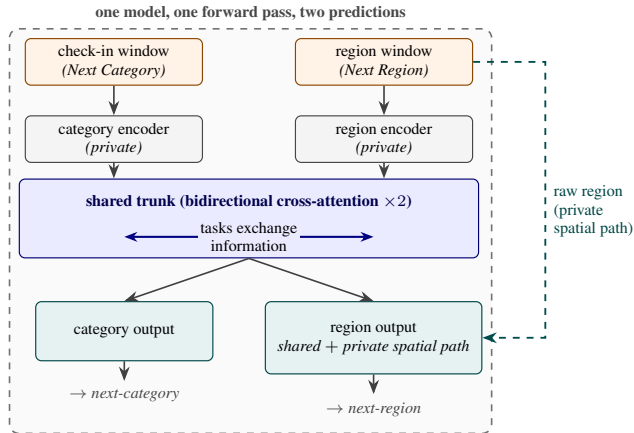
- ▶ **Semantic** — the category of the visited place, as an indicator;
- ▶ **Cyclical time** — time of day and day of week, through sine and cosine;
- ▶ **Elapsed time** — since the previous visit, since the first visit, the within-day gap, a first-visit indicator.

A design principle, in the chapter's own wording

Edges between consecutive visits run forward in time only. A target is predicted from a user's past, so the representation is built from the past alone — in training and at readout.

The architecture: sharing by exchange (1/2)

Two inputs, one model



Two inputs · 9 visits $\times$ 64 d each

- Next Category — per-visit vectors
- Next Region — same window

Heads

category: GRU · region: two STAN routes

Shared trunk

2 cross-attention blocks
4 heads · width 256

One shared module, no shared hidden layers: the two tasks EXCHANGE, they do not MERGE.

The architecture: sharing by exchange (2/2)

- ▶ **The private spatial path** — bypasses the shared trunk, feeds the region output only; **the category task never touches it**;
- ▶ **Fixed loss: 0.5 category + 0.5 region**, by design;
- ▶ **Logit adjustment** on category ($\tau = 0.5$, training only) — **the dedicated model receives the same**.

The joint-best convention · one saved model per fold

$$S_{\text{joint}} = \sqrt{\text{MacroF1} \times \text{Acc@10}}$$

The position on the shared trunk

The evidence does not separate the contributions of the shared trunk and the private spatial path. It does not establish that sharing helps, and it does not rule it out. The claim I make is about the design.

The protocol, in four steps

1 · the unit of data

- ▶ **User-disjoint five-fold cross-validation** — every window from one user stays in one fold; **the repair of the limitation declared in Chapter 3**;
- ▶ stratified by the **next-category label**, the seven classes being imbalanced;
- ▶ **sliding windows, stride 1** — nine visits in, the next visit is the target;
- ▶ the **held-out fold provides the validation data** — no third split is reserved. *Limit 2, later in this section, comes from this line.*

The protocol, in four steps

2 · what is measured

- ▶ **Category: macro-F1** — the mean of the per-category F1; reference point the majority-class floor, **5.7 to 7.3**;
- ▶ **Region: Acc@10** — the model returns a Top 10; we measure how often the true region is in it. *It does not separate first from tenth*, and a region unseen in training counts as an error;
- ▶ **Reference points for region** — the **dedicated model** and the **Markov-1 floor**, **51 to 72**.

The protocol, in four steps

3 · what is compared

- ▶ Joint model against the **dedicated single-task models**;
- ▶ Same representation, same windows, **same folds**;
- ▶ **A seed** = one full five-fold repetition: initialization *and* user partition;
- ▶ Four seeds {0, 1, 7, 100} × five folds = **20 fitted models** (60 h of GPU time A40);
- ▶ The test compares **four numbers**: one mean per seed.

The protocol, in four steps

4 · how it is decided

A claimed gain and a claimed match require different tests

- ▶ Analysis plan written **before any result was read**;
- ▶ **Superiority** → next category **only**;
- ▶ **Non-inferiority** → next region **only** — TOST (Lakens, 2017), $\delta = 2$ pp, fixed in advance;
- ▶ Paired t · 90% CI · **Holm** (1979) across the six datasets, **per task**;

Two consequences: a region gain is a secondary result, outside the plan; an unresolved category difference is not a match.

*The statistical evaluation protocol was later refined based on the literature.

Result 1: the representation, at every dataset

Next category, single-task model, seed 0 — only the input changes

Dataset	Check-in level	Place level	Δ
AL	30.77 ± 1.16	29.15 ± 0.84	+1.62
AZ	34.51 ± 1.15	31.93 ± 1.08	+2.58
Istanbul	35.35 ± 0.89	29.07 ± 0.63	+6.29
FL	37.36 ± 0.42	37.13 ± 0.41	+0.23
CA	35.62 ± 0.47	34.74 ± 0.41	+0.88
TX	36.32 ± 0.51	35.33 ± 0.48	+0.99

seed 0, five matched folds; $\pm$ is the fold sd.

Only the input changes — same target, model, training configuration, folds, windows, epoch budget, logit adjustment.

Result 2: one model, two tasks

Next category (macro-F1)

	POI-RGNN	Ded.	Joint
AL	23.80	30.77	<u>30.59</u>
AZ	27.64	34.57	34.57
IST	30.12	<u>35.34</u>	35.42
FL	34.49	<u>37.35</u>	37.55 ▲
CA	31.78	35.63	35.63
TX	33.03	36.33	<u>36.19</u>

Next region (Acc@10)

	HMT-GRN	ReHDM	STAN	Ded.	Joint
AL	57.05	65.38	60.72	70.12	<u>69.24</u> ≈
AZ	43.70	53.00	49.86	59.48	<u>59.04</u> ≈
IST	60.4	69.33	61.86	75.16	<u>75.08</u> ≈
FL	63.74	64.49	72.99	76.69	<u>76.54</u> ≈
CA	49.61	50.26 [‡]	58.52 [†]	<u>63.48</u>	64.54 ▲
TX	53.85	48.81 [‡]	61.67 [†]	<u>64.94</u>	66.15 ▲

- Above every external system we ran, on both tasks — at least **3.06** macro-F1 on next category;
- it exceeds the **first-order Markov floor** by **+4.1 to +10.0** at every dataset, while **HMT-GRN stays below that floor at all six**; STAN at four, ReHDM at three.

HMT-GRN: same data, folds and seeds (primary comparison). [†]STAN: our re-implementation, output adapted to regions; TX 4/5, CA 2/5 folds, seed 0. [‡]ReHDM: its own published protocol, single seed on CA/TX. Ties in bold on both.

The verdict, dataset by dataset

	Next region (Acc@10)		Next category (macro-F1)	
	Δ	90% CI	Δ	90% CI
Alabama	-0.87	[-1.00, -0.75]	-0.19	[-0.33, -0.04]
Arizona	-0.44	[-0.62, -0.25]	-0.00	[-0.04, +0.03]
Istanbul	-0.08	[-0.16, -0.002]	+0.08	[+0.01, +0.15]
Florida	-0.16	[-0.19, -0.13]	+0.19 ▲	[+0.14, +0.25]
California	+1.06 ▲	[+1.03, +1.08]	-0.00	[-0.03, +0.02]
Texas	+1.21 ▲	[+1.13, +1.29]	-0.13	[-0.19, -0.08]

▲ Outperforms the dedicated model, Holm-corrected — region: Texas ($p = 0.00013$) · California ($p < 10^{-4}$), 20 of 20 folds; category: Florida ($p = 0.011$), 19 of 20. **The two region gains are secondary results, outside the registered plan.** **Region** — the other four stay within the two-point margin, and **all four are deficits, not ties**: every interval lies entirely below zero.

Category — the other five are **unresolved**; all six within half a point, read off the intervals. **The plan registered no equivalence margin on category.**

Limitations and trade-offs

THE TRADE

- ▶ Larger
AL 4.2 M vs 1.85 M — 2.3×
CA 5.2 M vs 2.80 M — 1.8×
- ▶ Operational, not computational
- ▶ Largest region deficit: **0.87 Acc@10** (Alabama)

FOUR LIMITS

1. Representation trained **once, over all places**
2. **No third split** — absolute scores are optimistic
3. **No mobility-aware service** built or evaluated
4. Each visit reads only its own past (*slide 30*)

Sumário

- 1 Abertura — a pergunta e o escopo
- 2 Fundamentos compartilhados
- 3 Multitask Learning for POI Category and Next-POI Prediction
- 4 ST-MTLNet: Spatio-Temporal POI Representations
- 5 A Check-in-Level Multitask Study of Next Category and Region
- 6 Conclusão Geral — a resposta condicional

Three studies, three layers

	Representation	Sharing topology	Protocol	What moved
MTLnet (Ch. 3)	one vector per place	hard sharing + FiLM	by sample · one seed	A null result , and three named suspects
ST-MTLNet (Ch. 4)	three encoders: space · time · category	<i>unchanged, by design</i>	<i>unchanged, by design</i>	The representation is the bottleneck
Check2HGI (Ch. 5)	one vector per visit	cross-attention · private spatial path	by user · four seeds	One model replaces two, on both tasks

The conditional answer

The answer

Multitask learning helps next-category and next-region prediction **under the final design and evaluation protocol developed in this dissertation**. The conclusions are study-specific; what carries across studies is the methodology and the conditions identified.

What it does not authorize

Not that multitask learning always helps: task relatedness and joint training were **not sufficient by themselves**.

Three conditions, and they do **not** carry the same weight of evidence:

- ▶ **Input representation:** established by controlled ablation (*Chapter 4 is that control*);
- ▶ **Architecture:** suggested, not isolated;
- ▶ **Scale:** **a possible condition, not an established cause.**

Contributions

Practical — what it delivers

- ▶ One artifact, two tasks — **operational, not computational**
- ▶ Check2HGI, published and **reusable**
- ▶ An **evaluation protocol**, released with the code

Scientific — what it establishes

- ▶ **Above every external system we ran, on both tasks**
at least **3.06** macro-F1 on next category
- ▶ **Baseline results for next region** — no fixed protocol existed
- ▶ **Check2HGI** — trained with **neither target**
- ▶ **Three conditions** for when multitask learning helps
representation **established** · architecture and scale **suggested**

Limitations

-
- | | |
|---------------------------------|---|
| 1 · Data vintage | Gowalla 2009–2011 · Istanbul 2012–2018 — a field-wide constraint |
| 2 · Taxonomy coarseness | seven top-level classes |
| 3 · Transductive representation | no unseen places or users without retraining |
| 4 · No next-place task | conclusions cover next category and next region only |
| 5 · Geographic coverage | outside the United States, one city |
| 6 · The task-pair confound | the pair changed together with the representation and the topology;
Chapter 4 is the fixed-pair control |
-

Future work

Input representation

An inductive Check2HGI — new places and users, no retraining
a hypergraph over sessions
attention-based graph encoders

Sharing topology

The exact next place — a third head, same representation
deeper sharing — one trunk, per-task heads
one training stage — representation and model together

Problem scale

Region count vs. data volume — the controlled experiment
newer traces · finer taxonomies
· new cities
the geographic size of a miss

Eleven items in the volume, each tied to a named limitation.

Closing

Does multitask learning help next-category and next-region prediction?

Under this design, one model predicts two properties of the next visit in one forward pass, and **replaces two dedicated models on both tasks**.

And what does the answer depend on?

The input representation, the sharing topology, and the scale of the problem.

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Extras

Fora do relógio. Uma pergunta por tela,
e toda entrada do índice é um link.

Se a pergunta for uma destas, o slide já existe

BO — Index

Seven families, one question per slide. Every entry is a link.

V19 • Rebuilt protocol (post-submission)

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B3 • Post-submission

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B4 • Chapters 3 and 4

B4-LEAK B4-DGI B4-1 B4-2 B4-3 B4-4 B4-5

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B6 • Document and scope

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Migrado da trilha principal

B-GEO

B7 • How Check2HGI and the joint model work

B7-1 B7-2 B7-3 B7-4 B7-5 B7-6

série de reserva • fora da contagem principal

The rebuilt protocol

1. V19-1 · Different protocol: sealed test, drawn partition, margin five times stricter

What changed	Delivered	Rebuilt	Why
Test set	read on the fold that also selects the epoch	sealed, read once — 81 reads this campaign, versioned ledger	a number read on the fold that selected it is optimistic
Partition	one integer moved both the split and the initialization	two, drawn apart — one seeds the split, one the initialization	the four seeds already spanned four partitions; what was missing was separating partition from initialization
Seeds	fixed count	adaptive ladder — TX 3 · FL, AZ, CA 4 · AL, IST 6	none was arbitrated — the ladder runs until it decides
Margin	$\delta = 2$ pp	$\delta = 0.4$ pp — four possible verdicts	registered before any reading

Next category, sealed test

2· V19-2 · Different protocol: sealed test, drawn partition, margin five times stricter

Dataset	Users	P	Δ	90% CI	Verdict
Alabama	1,101	6	-0.377	$[-0.584 \cdot -0.177]$	inconclusive
Arizona	2,136	4	-0.115	$[-0.224 \cdot -0.012]$	non-inferior
Florida	10,622	4	+0.136	$[+0.097 \cdot +0.176]$	superior
Istanbul	14,530	6	+0.155	$[+0.076 \cdot +0.232]$	superior
California	20,667	4	+0.035	$[+0.015 \cdot +0.058]$	non-inferior
Texas	22,592	3	-0.007	$[-0.030 \cdot +0.017]$	non-inferior

Both at once: region non-inferior and category no worse — four of the six

Both protocols agree the category effect is small: 3 of 6 carry a sign, 2 carry none, 1 disagrees.

Texas category is the one disagreement: -0.13 excluding zero, -0.007 including it — consistent with what removing the selection bias produces.

Next region, sealed test

3. V19-3 · Different protocol: sealed test, drawn partition, margin five times stricter

Dataset	Users	P	Δ	90% CI	Verdict
Alabama	1,101	6	−0.925	[−1.099 · −0.767]	inferior
Arizona	2,136	4	−0.572	[−0.719 · −0.430]	inferior
Florida	10,622	4	−0.219	[−0.257 · −0.179]	non-inferior
Istanbul	14,530	6	−0.122	[−0.181 · −0.074]	non-inferior
California	20,667	4	+0.961	[+0.908 · +1.029]	superior
Texas	22,592	3	+0.893	[+0.853 · +0.934]	superior

Both at once: region non-inferior and category no worse — four of the six

pós-submissão — não consta em nenhum dos dois volumes. mtlcheck a6639c0a.

Capacity in MTL

1. B-KARPATHY · An open problem the field cannot design in advance

- ▶ Karpathy (2019), on designing a multitask network: *“how much feature sharing is there”, “tasks fight for the same shared capacity”, “there’s finite capacity to go around... I don’t really have language to describe how to correctly allocate capacity to tasks”*.
- ▶ Standley et al. (ICML 2020): the task-affinity matrix, which asks the same question empirically.
- ▶ PCGrad and GradNorm, which that discussion names, are **already in this dissertation** (Chapter 2).
- ▶ **The point:** the capacity control in **B-P1** is exactly the measurement this literature says the field does not know how to design in advance. Running it is the answer to an open problem, not a wound.

Karpathy (2019), palestra; Standley et al., ICML 2020; Cap. 2, §2.3. contexto de literatura, fora dos dois volumes.

Capacity control

1. B-P1 · Matched capacity removes the advantage at California; Texas is unresolved

Give the dedicated region model the joint model's entire parameter budget. Seed 0, five folds.

Dataset	ded. (narrow)	ded. (matched)	joint	joint – matched	p	unanimous
California	63.446	64.931	64.503	– 0.428	0.0082	5/5
Texas	64.951	66.330	66.117	–0.214	0.1162	4/5

- ▶ Width curve **saturates early**: **57 percent** of the budget already reaches the same level (352 → 528: +0.021, p 0.40).
- ▶ What survives: one model, both predictions, one forward pass, no measured cost on either task.
- ▶ **Category is untouched**: 6.5 times the parameters scores **lower** by 0.53 macro-F1 (p 0.0011).
- ▶ Volumes: the only capacity control is **Appendix G** — and it covers **next category**
- ▶ Its stated limit: width varied, representation held fixed; the joint model not decomposed

Concatenation control

1. B-Q13 · Stands: representation over architecture. Falls: which part carries the gain

The deposited sentence is wrong in its direction. There is a written errata. Control redone on the scale of Table 9 — one seed, five folds, only the input changes.

Dataset	gap (place → check-in)	concatenation gain	share of the gap
Alabama	+1.56	+1.73 ($p = 0.003$)	the whole gap
Arizona	+2.50	+1.70 ($p < 0.001$)	68 percent
Florida	+0.21	+1.02 ($p < 0.001$)	past it

- ▶ AZ: check-in-level leads by **0.80** • AL: within **0.18** • FL: concatenation leads by **0.81** (p 0.03 / 0.53 / 0.001).
- ▶ **Stands:** input representation dominates architecture. **Falls:** which part of it carries the gain.
- ▶ Fidelity: AL fold by fold; AZ/FL match the mean within 0.07/0.03, **not** fold by fold — **open**.

Q13_concatenation_control.md (2026-08-16); errata escrita. pós-submissão: não consta nos dois volumes.

Capacity confound

1. B-Q14 · The paper lists it; the dissertation does not

It should not have gone out. It returns as an errata, and it now carries the measurement the paper said was missing.

- ▶ **Submitted paper, p. 9, limit 4 of 5:** the region advantage at Texas and California “*is therefore confounded with capacity*”, and the capacity-matched control “*has not been run*”.
- ▶ **Main volume, p. 85:** “*Four limits qualify these results*” — capacity on the region axis is not one of them.
- ▶ What it **does** carry: the parameter cost on p. 73 (about **4.2 million** at Alabama against **1.1 million** for the two dedicated combined) and an attribution on p. 84 crediting neither trunk nor transfer.
- ▶ The only capacity control in either volume is **Appendix G of the supplement: next category only**.

artigo submetido 07_discussion.tex:105–109, p. 9; volume principal p. 85 e p. 73;
Apêndice G do suplemento, p. 24–26.

Checkpoint convention

2· B1-3 · We report the convention that yields fewer improvements

Reported convention: each dedicated model at its own task's best epoch; the joint model at the epoch its joint validation score selects — both tasks from the one saved model.

The alternative (each task at its own best epoch) is more favorable to the joint model:

- ▶ at most **0.23 macro-F1** and **0.93 Acc@10** at any one seed;
- ▶ over the four seeds, **+0.03 to +0.17** category and **+0.19 to +0.90** region;
- ▶ **four further category cells and two further region cells** turning into improvements that survive the same Holm correction.

That is exactly why it is not the one reported. One checkpoint per fold is what a deployed system can serve.

Direction of the four

2· B1-1 · All four intervals lie entirely below zero

All four are deficits. All four intervals lie entirely below zero.

Dataset	Δ Acc@10	90% CI
Alabama	−0.87	−1.00 to −0.75
Arizona	−0.44	−0.62 to −0.25
Florida	−0.16	−0.19 to −0.13
Istanbul	−0.08	−0.16 to −0.002

- ▶ Reverse-direction test, post hoc over the same six comparisons and corrected across them, resolves three of the four. **Istanbul is not resolved.**
- ▶ Each clears the registered two-point margin. **None of them is a tie.**

No category margin

2· B1-2 · Category has no margin; the intervals carry the bound

Superiority registered on next category, **non-inferiority** on next region. **No equivalence margin on the category axis.**

Dataset	Δ macro-F1	90% CI	favors
Istanbul	+0.08	+0.01 to +0.15	joint model
Arizona	−0.00	−0.04 to +0.03	no direction
California	−0.00	−0.03 to +0.02	no direction
Texas	−0.13	−0.19 to −0.08	dedicated
Alabama	−0.19	−0.33 to −0.04	dedicated

- ▶ Florida (+0.19) is the one cell that survives Holm.
- ▶ The other five are **unresolved**; the widest interval reaches 0.34 from zero, at Alabama — **equivalent to zero within half a point.**

Epoch selection

2· B2-3 · Every absolute score here is optimistic

Yes — the second of the four declared limits, in the chapter's own words: *“epoch selection consults the fold that the score is then read on, so every absolute score reported here is optimistic”.*

Why the **comparison** is affected far less, stated rather than assumed:

- ▶ same selection rule for both models on the same folds, each selected on its own validation objective;
- ▶ for category the **dedicated** model receives the **wider search**: batch size at all six datasets and learning rate at four, against a joint-model search covering four of six;
- ▶ both sides of the **region** comparison run one fixed configuration, so that mitigation does not apply there;
- ▶ the chapter closes it: *“It does not follow that the bias cancels exactly”.*

Transductive by design

2· B2-1 · A per-fold rebuild moves at most 0,33 Acc@10 — three datasets, one seed

- ▶ The representation objective **never reads the next-category or next-region targets**.
- ▶ Control: a fresh representation built **per fold, from that fold's training users only**. Three datasets, one seed. Differences: **−0.33 to +0.01 Acc@10 (region), 0.00 to +0.29 macro-F1 (category)**.
- ▶ Declared limit of that control: a training-only graph has no visit vectors for validation users, so the category comparison used one vector per place and kept only windows whose input places occurred in training. **Those windows cover 67 to 87 percent of the validation data**; per-visit information and unseen places are not covered.
- ▶ Forecast evaluation is **user-disjoint**; representation learning is **transductive** with respect to the graph, and the chapter says so.

Beating the literature

2. B6-5 · Yes on category; on region the floor is the reference

- ▶ **Next category — the comparison is clean.** POI-RGNN is **native to the task**, re-implemented from its published architecture, and **above the tuned Markov-K floor at all six**. The chapter's own sentence: *"the joint model stands **at least 3.06 points** above the strongest external baseline at every dataset."*
- ▶ **And the chapter uses that as an argument, not a victory:** representation **+0.23 to +6.29**, the gap over the literature **at least 3.06**, one model against two **half a point** — **"the smallest of the three effects at work here."**
- ▶ **Next region — more careful, and here is why.** No published model targets this exact task: the nearest, DRRGNN, predicts region over **regions discovered per person**, not a fixed citywide partition. And the **Markov-1 floor is above HMT-GRN at all six, STAN at four** — so the chapter treats **the floor, not the external systems**, as the reference.
- ▶ **The caveat for both.** The externals run on their own embeddings, so this margin **includes the representation advantage**. The comparison that decides the thesis is the **Dedicated** column.

How each baseline ran (1 of 2)

2· B6-6a · Next category

Sistema	Tarefa	Como rodou	A ressalva
POI-RGNN	categoria	re-implemented from its published architecture and hyperparameters , on our folds	nenhuma — é nativo da tarefa e está acima do piso Markov-K nos seis
Markov	ambas	best order per dataset (categoria); Markov-1 sobre transições de região, sob as nossas janelas e partições	não é figurante: é a referência que o capítulo declara

Por que a categoria dele não conta: o **CSLSL** prediz categoria, mas a usa como **degrau para o lugar** — a saída principal é a localização. Cap. 5, §5.2; 02_related.tex.

How each baseline ran (2 of 2)

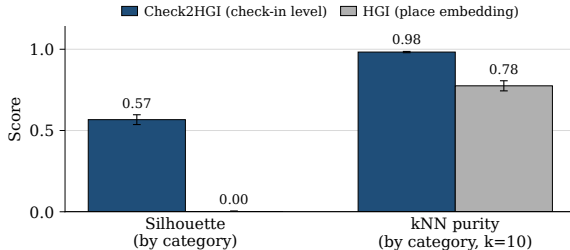
2· B6-6b · No published model treats region as an end target

Sistema	Tarefa	Como rodou	A ressalva
HMT-GRN	região	mesmos dados, partições e inicializações; estrutura multitarefa preservada ; prior de transição por partição	<i>“not a reproduction of the complete published system”</i> — o que saiu serve à busca do lugar . Removido: componentes de grafo e busca em feixe hierárquica — <i>servem ao próximo lugar</i>
STAN	região	mesmas partições, mas embeddings e sequências próprios ; saída adaptada para ranquear regiões	partições parciais: TX 4/5, CA 2/5, semente O. Não adaptado a categoria: casamento por distância não se aplica a categorias
ReHDM	região	sob o protocolo publicado dele	não medido nas nossas janelas nem partições; TX e CA com uma semente

Por que nenhum é alvo final: nenhum modelo publicado ataca este alvo; o mais próximo, o **DRRGNN**, prediz região como fim, mas sobre **regiões descobertas por pessoa**. Cap. 5, §5.2 e §5.5.4; 05_setup.tex.

Geometry of the vectors

2. B-GEO · The vectors separate categories — regions they do not



- **Silhouette by category** — how tight and how well separated the seven labeled groups are, on a scale of -1 to 1 ;
- **Nearest-neighbor category purity** ($k = 10$) — the share of a vector's ten nearest neighbors that carry its own category. *Both averaged over the five U.S. states; Istanbul is not in this average. These measures characterize the representation family, not the exact configuration evaluated later.*
The same geometry does not separate regions: the benefit is category-only.

Task pair

3. U6 · No clean ablation separates it; the fixed-pair control bounds it

- ▶ Limitation 6, p. 90: no controlled ablation separates the change of representation and topology from the change of task pair.
- ▶ The ablation that would separate them — **static category classification under the check-in-level representation** — *“is not clean under that representation: the category of the visited place is an input feature of a check-in node, so that target would be partly readable from its own input”*. **The confound is bounded by the fixed-pair control rather than removed.**
- ▶ **Chapter 4 is the fixed-pair control:** same architecture, same task pair, only the input moves.
- ▶ **Why it does not overturn the thesis:** the thesis is conditional by construction (p. 89), and Chapter 4 carries the claim that the representation is the dominant factor.
- ▶ Future work tied 1:1 to the limitation, p. 91: a static target the representation does not carry as an input feature.

Transfer or architecture

3. Q8 · The claim is the design, not transfer between the tasks

- ▶ Chapter 5, p. 84: *“The evidence here does not separate their contributions”*. The surviving claim is about the **design**: shared representation and private spatial path together, above two dedicated models at the two datasets with the largest region vocabularies.
- ▶ **A one-fold screen** (seed 0, three arms; one number per arm, so it detects only a large effect): at California the region advantage survives severing the trunk (-0.099) and deleting the category task (-0.077); at Texas it survives severing the trunk (-0.120). All under 0.15 point.
- ▶ **The five-fold trunk ablation at those two datasets does not exist**. Five-fold ablations ran only at Alabama and Florida.
- ▶ **Post-submission**, a capacity-matched control answers the other half: at California the advantage **does not survive matched capacity** ($p = 0,0082$); at Texas it is **unresolved** ($p = 0,1162$).

Label channel, Ch. 3 × 4

3. B4-LEAK · Exact lookup in Chapter 4; one-hop average in Chapter 3

- ▶ **Chapter 4:** the venue-type feature maps **one to one** onto the seven top-level categories, 284 to 365 distinct values per state, **not one ambiguous** — an **exact lookup**.
- ▶ **Chapter 3:** the input feature is the **average of its neighbors' categories**, own one-hot excluded by construction. **The exclusion holds exactly, and it does not close the channel** — each neighbor's feature is itself a neighborhood average, and the place belongs to those neighborhoods.
- ▶ Measured: own category returns at **mean weight 0.10** against **total own-category weight 0.39**; removing it lowers a probe of the own category from **0.46 to 0.30 macro-F1**, against a **majority-class floor of 0.07**. Confirmed by causal intervention.
- ▶ **Does it invalidate the chapters? No.** The **sequential** task of both never has the target in its input, and that is where the arc's conclusions come from.

next category (Def. 2.7) • Chs. 3/4 report **per-category F1** • auditoria de código (DGI-leak-audit); Cap. 4, p. 52.
não chegou a nenhum dos dois volumes.

Forward edges

3. B2-4 · Forward-only by design; still transductive by construction

- ▶ **In the delivered text** the direction is a **design decision**, stated as the fourth of the four limits (p. 85) and in the method (p. 26): each visit node draws on the visits that precede it, in training and at readout, *“which is what keeps a node from carrying a feature of the target it is used to predict”*.
- ▶ **In the repository**, the generation that produced every delivered cell is the one in which that channel is closed. Closing it was worth **28.63 macro-F1 at Alabama** on the category axis; the region axis moved by under two points.
- ▶ **What the closure does not buy**: the representation is still trained **once over the whole graph** — transductive by construction. The forecast splits stay user-disjoint.
- ▶ Neither volume reports a before-and-after of this channel; this slide does not present one as a result.
- ▶ **Supplement, Table 4, p. 18**: a fourth ground over-declared — the fix **removes the row**, it does not write an errata

Cap. 2, p. 26; Cap. 5, p. 85 (limite 4). O valor 28,63: CLAUDE.md §0.2 e NEW_VERSION.md §8.
pós-submissão / repositório: não consta em nenhum dos dois volumes.

Two declared qualifiers

3. B4-3 · Best-of-two per row, and not width-matched

- ▶ **The range is best-of-two per row, and saying so is the answer.** Category gains of 20.2 to 22.0 points per state count, for each category-state cell, the better of SIREN and Sphere2Vec-M.
- ▶ Read **by isolated variant**, SIREN alone at Texas averages +17.89, outside the announced range.
- ▶ On the sequential task the same rule gives **15 of 21** category-state combinations to the variants, with one technical tie at Outdoors in Florida (baseline mean above the best variant by 0.02, inside one standard deviation).
- ▶ The chapter states the pattern as an observation: SIREN stands out in Florida and California, Sphere2Vec-M more often in Texas. **No universally better spatial encoder, and no rule for choosing one without seeing the evaluation.**
- ▶ Shared capacity **unchanged at 256** — what differs is **input width**, 192 against 64
- ▶ The equal-dimension control: requested (Ch. 4 p. 61, Ch. 6 p. 87), **never run**

next category (Def. 2.7) • Ch. 4 reports **per-category F1** • Cap. 4, Tabela 6, p. 63; Tabela 7, p. 65; variante isolada em `judge_feedback.md:11`.

Bottom-up hierarchy

3. B7-3 · Four steps, and one deliberate gradient cut

1. **Two check-in graph-convolution layers** over the succession edges, residual update. Output: one 64-dimensional vector per visit.
2. **Pool visits at their place** with four attention heads, one learned query shared across places, keys and values from that place's visits.
3. **Add the spatial place neighborhood:** the pooled place representation combined with a trainable place table initialized from a pretrained one, then one weighted graph convolution over the Delaunay place graph. **The pooled place representation is detached on this route**, so the place-region and region-city objectives cannot rewrite the check-in encoder through the spatial branch.
4. **Region and city:** a second four-head pooling over the places of a region, a 64 to 64 convolution between adjacent region polygons, then an **area-weighted city summary**. The city vector serves the highest objective and is **not** an input to the joint model.
 - ▶ Check-in node: **15 values** — coordinates are **not** among them
 - ▶ Succession edge: **one direction**, 1 h decay · category co-occurrence is **not** an edge

Apêndice E do volume principal, §§ “Step 1” a “Step 4”, pp. 110-112; gradiente interrompido em p. 112.

Cross-attention pairing

3. B7-6 · Training batches pair rows at random; validation rows are record-aligned

- ▶ **During training: yes.** The category and region loaders use the **same user-disjoint fold** and **shuffle independently**, so their batch rows have compatible shapes but *“need not describe the same user or window in one optimizer step; cross-attention operates on that random pairing”*. The shorter loader cycles until the longer one is exhausted.
- ▶ **At validation: no.** *“Validation rows are record-aligned.”*
- ▶ The appendix calls it what it is: *“This operational detail is unusual, but it is part of the reported training protocol.”*
- ▶ Objective: **fixed 0.50 / 0.50** cross-entropy, logit adjustment $\tau = 0.5$ on the category term at train time only. AdamW, three parameter groups, one backward pass. Fifty epochs, five user-disjoint folds, seeds {0, 1, 7, 100}. **The checkpoint maximizes the geometric mean of category macro-F1 and region Acc@10.**

Not hard sharing

3. B7-4 · Private encoders and heads; only activations meet

- ▶ **Private encoders, same shape, different parameters:** each history goes through $64 \rightarrow 256 \rightarrow 256$, ReLU and layer normalization after every transformation, dropout 0.1 after the first two. Equal tensor widths do not imply equal meaning: a check-in vector describes one visit in a trajectory, a region vector a geographic area after spatial aggregation.
- ▶ **Two bidirectional cross-attention blocks.** In each block Next Category queries Next Region first; Next Region then queries the already updated Next Category. Four heads, padding positions excluded, each direction with its own projections, residual connections, normalizations and a $256 \rightarrow 256 \rightarrow 256$ GELU feed-forward network.
- ▶ **Jointly optimized, but the directional projections are not tied:** *“The model therefore differs from classical hard parameter sharing: the tasks keep private encoders and heads, while their activations meet in a fixed trainable interaction module that receives gradients from both losses.”*

The region tower

3. B7-5 · Two routes, and one prior fixed at zero

- ▶ **Category head:** a four-layer unidirectional GRU, width 256, reading the category-context sequence; the top-layer state at the last valid position, then layer normalization, dropout, seven logits.
- ▶ **Region head keeps two routes on purpose:**
 - ▶ **private tower:** the raw 9×64 region history, spatio-temporal attention, four heads, dropout 0.3;
 - ▶ **shared-context tower:** the 9×256 cross-attention sequence, eight heads, dropout 0.1;
 - ▶ each returns one 128-dimensional feature, fused as $f_R = f_{\text{priv}} + \beta \cdot W_{\text{shr}} f_{\text{shr}}$, with β trainable, initialized at 0.1.
- ▶ **Category context helps region prediction without removing the direct spatial sequence model.**
- ▶ **One term is inactive here:** an additive region-transition prior, scalar weight fixed at zero and not trained — it reaches neither the logits nor the gradients. **Region prediction depends only on the two towers.**

Shared origin

3. Q5 · Declared without a number; the only bound is architectural

- ▶ **Declared, without a number**, in Chapter 2, p. 27: the joint model reads two tables exported from the same check-in-level representation, and *“The two tables share an origin by construction, so they are not independent views”*.
- ▶ The only **quantified** boundary is architectural: on the spatial route the pooled place representation is **detached** (Appendix E of the main volume, p. 112).
- ▶ That bounds gradient flow **inside representation training**. It does not quantify the **information overlap** between the two exported tables.
- ▶ **Why it does not overturn the thesis**: the comparison that carries the conclusions is paired and internal, under the same representation, windows and folds. Shared origin bears on the mechanistic reading p. 84 already leaves open, not on the validity of the measured difference.
- ▶ What would close it: **one measured overlap** between the two windows on the same sequence

The unit of the test

3. B1-4 · $n = 4$; the exact Wilcoxon floors at 0,0625

- ▶ 4 seeds $\times$ 5 folds = **20 fitted models** per configuration. the test compares **four numbers**: one mean per seed.
- ▶ **Primary**: paired t on the four per-seed means, with the 90% CI — folds inside a seed are not independent.
- ▶ **Registered**: paired Wilcoxon signed-rank over the 20 matched fold differences. **Reported alongside, and it agrees.**
- ▶ At the seed-level footing the exact one-sided Wilcoxon cannot fall below **0.0625**, whatever the effect size. That is why the t carries the verdict.
- ▶ The departure is stated in the chapter and both tests ship in the code release.

The region floor

3· B6-4 · Computed under our own windows — the chapter declines one explanation

- ▶ HMT-GRN falls below the floor at **all six** datasets, the ReHDM reference at **three**, STAN at **four**.
- ▶ The floor is computed under **our own sliding windows and folds**, advancing one visit at a time, *“so the region of the last visit is a strong predictor of the next one”*. At Alabama the target is the last visited region in **32.9%** of windows.
- ▶ They do not meet the floor on equal terms: HMT-GRN on the same data, folds and inits; STAN on the same folds, own representations; ReHDM under its own published protocol.
- ▶ *“We do not claim a single explanation. We treat the floor, not the external systems, as the reference the region task has to clear.”* The joint model exceeds it by **4.1 to 10.0** points at all six datasets.

The Resumo defect

3. B6-1 · One sentence, isolated, corrected in the source

	what it says
Resumo (delivered)	superiority on next category <i>“em todos os conjuntos”</i>
English Abstract, §2.5, Ch. 5, Ch. 6	superiority at one dataset
The delivered result	Florida only , +0.19, Holm p 0.011

- ▶ The defect was **isolated to the Portuguese Resumo**. Every other surface of the document already said one dataset.
- ▶ Corrected in the source for the final deposit, with a registered errata. The delivered PDF is kept **unchanged** as the record of what the banca received, and it differs from the source in this one sentence.

Who is in the test

3. B6-3 · The user column is the raw corpus, not the test

Table 8 column	population
check-ins • users • POIs windows	raw corpus after the minimum-length filter (ten check-ins), stride 1

- ▶ Verified by direct count on the raw files: Alabama 113,846 check-ins, **3,858 users**, 11,848 places; Arizona 236,450, **7,869**, 20,666 — exactly the printed values.
- ▶ Counted on the representation actually used, the users **present in the pooled predictions** are **1,101 (AL)**, **2,136 (AZ)** and **14,530 (Istanbul)**. The window column agrees exactly in all three (96,326 • 200,895 • 271,666).
- ▶ Arithmetic closes both ways at Alabama: the 1,101 qualifying users hold 106,235 of the 113,846 check-ins, leaving 7,611 for the 2,757 who do not qualify, about 2.8 each.
- ▶ **A caption clause fixes this without touching a number.**

Tabela 8, p. 75 do volume principal; reconciliação em NEW_VERSION.md §10.1. **a contagem pós-filtro é pós-submissão: não consta nos dois volumes.**

Travel, by task

3. B4-2 · Category gains; the sequential task loses

Travel, Florida	MTLnet	ST-MTLNet (SIREN)
category (Table 6)	45.49 ± 1.20	64.89 ± 1.20
next category (Table 7)	64.47 ± 1.02	45.00 ± 1.10

- ▶ California repeats it on the sequential task: 46.05 ± 0.84 against 36.94 ± 1.70 and 37.82 ± 1.04 . **Texas does not:** the variants lead (34.26 ± 0.90 against 29.71 ± 1.32).
- ▶ The chapter's reading, p. 64: Travel “*tends to involve sparser movements between distant regions*”, where the DGI graph topology “*may be more efficient for preserving relationships between geographically distant POIs*”.
- ▶ Declared as a limitation (p. 66), with the fact that the three encoders are used jointly and none is isolated.

next category (Def. 2.7) • Ch. 4 reports **per-category F1** • Cap. 4, Tabela 6, p. 63; Tabela 7, p. 65; leitura p. 64; limitação p. 66.

Fitting order

3. B7-1 · Fitted first and frozen — the forecast labels never reach it

1. validate the records, order each user's visits by time, map each place to a polygon;
 2. build temporal, place and region graphs linked by the check-in / place / region / city hierarchy;
 3. train Check2HGI, export separate **64-dimensional** check-in and region tables;
 4. build stride-one windows, nine observed visits, the tenth as target;
 5. train one joint model on the two representation sequences.
- ▶ **Check2HGI is fitted first; its exported tables stay fixed during supervised training.** The joint model does not rebuild the graph or update the representation encoder. **Check2HGI never receives the two forecast labels.**
 - ▶ Learning signal: **bilinear discriminators at three hierarchy boundaries** (check-in to place, place to region, region to city), each separating a true pairing from a corrupted one.
 - ▶ Two auxiliary terms: masked-place reconstruction (15 percent hidden) and a place-table anchor. Full-batch Adam, 500 epochs, learning rate 1e-3, clip norm 0.9.

Future work, in full

4. B-FUTURE · Eleven, grouped by what they touch

Representation	an inductive variant — unseen places and users, no retraining (L3); a hypergraph formulation, one edge per session (L3); an ablation of the pretrained place-table coupling (L3); Check2HGI inside other mobility architectures
Joint model	the exact next place as a third target — own head, or a cascade (L4); a shared trunk with MMoE , as an alternative to cross-attention; one end-to-end stage — representation and joint model trained together
Evidence	newer, denser traces (L1) · finer taxonomies (L2) · cities outside the United States (L5); a static target the representation does not carry as a feature (L6); a controlled experiment separating region count from data volume

The two-point margin

4. U8 · Registered before results; Alabama would fail one point

- ▶ The justification is a declared judgment: a service acts on which region will be busy, not on a single rank position, and “*A two-point change in Acc@10 is below the level at which this service would behave differently*” (p. 77).
- ▶ The empirical support that does exist is the dispersion: the sd of the paired difference across the four user partitions runs **0.02 to 0.16**, and the intervals at **Istanbul, Arizona and Florida** would support a **one-point** margin. **Alabama’s would not.**
- ▶ **Why it does not overturn the thesis:** the margin was registered **before any result was read** (p. 76), and the four cells inside it clear it comfortably, the largest deficit being **0.87** (p. 82). Under a one-point margin three of the four datasets still hold by their own intervals, and the document names the exception.

Trunk at TX and CA

4. U1 · The five-fold trunk ablation at Texas and California does not exist

- ▶ What exists: the one-fold screen of **Q8**, where every arm moves under **0.15** point.
- ▶ The five-fold ablation at those two datasets **does not exist**. Five-fold arms were run at **Alabama** (dcat -0.015 / dreg -0.138 , and dcat -0.154 / dreg -0.004) and **Florida** (dcat $+0.002$ / dreg $+0.026$).
- ▶ **Why it does not overturn the thesis**: the thesis does not claim the trunk carries the result. The claim on p. 84 is about the full design, and the same page states that the evidence does not separate the two parts. The screen had power for the hypothesis that the trunk carries the two points and did not confirm it — which **supports** the cautious wording rather than contradicting it.

region_1fold_triage/FINDING.md; SWEEP_PLAN.md:275-277, :285-290; Cap. 5, p. 84.
repositório: não consta nos dois volumes.

Gradient cosine

4. U4 · Four of six measured; the two largest label spaces are not

- ▶ Appendix D of the main volume covers **four of the six** datasets, and says so: *“Texas and California are not measured here”* — precisely the two where the joint model outperforms on region. The appendix names that too: the diagnosis *“leaves the largest label spaces untested”*.
- ▶ It also limits itself by architecture: *“Nothing here says the gradients stay orthogonal in a model that shares more of its depth, couples the tasks in a cascade, or shares an output layer”*.
- ▶ At the four measured datasets, equivalence to zero holds within a ± 0.05 margin, with every mean inside it and **99.6 percent** of individual epochs (Figure 8, Table 11).
- ▶ **Why it does not overturn the thesis:** the appendix explains why a gradient balancer had nothing to balance, supported independently by the screen of nineteen balancers at two datasets (p. 70). No conclusion of Chapter 5 depends on the cosine at Texas or California.

How far the miss

4. U5 · Not retained by the evaluation path

- ▶ Chapter 5, p. 85: *“Where the shortlist misses, the geographic size of the error is the quantity that would matter to such a service, and measuring it requires the per-visit predictions that the evaluation path does not retain, so it is left to future work”.*
- ▶ The service framing is explicitly motivation, not result, and is the **third** of the four declared limits: *“we do not build or evaluate a mobility-aware service”.*
- ▶ **Why it does not overturn the thesis:** no claim in the document is about service performance. The shortlist reading on p. 84 comes with the number that supports it, the Acc@10 of Table 10 itself: **California 64.54 percent** in ten tracts out of 8,501; **Texas 66.15 percent** in ten out of 6,553.

Other models

4. U7 · Not evaluated, and transductive by construction

- ▶ Chapter 6, p. 88: the result *“also supports testing Check2HGI in other mobility prediction architectures, although its benefit in those architectures has not yet been evaluated”*.
- ▶ Limitation 3, p. 90: the representation is **transductive**, trained on each dataset’s check-in graph, *“so it cannot represent unseen places or users without retraining”*.
- ▶ **Why it does not overturn the thesis:** every comparison that carries the thesis holds the consuming model fixed and varies only the representation. Table 9 is exactly that design: same single-task model, same training configuration, same folds, same sliding windows, same epoch budget; only the input changes.

Two extractions

4· B4-4 · Every record of the earlier extraction reappears in the current

	users	POIs	check-ins
Chapters 3 and 4	20,301	65,009	990,518
Chapter 5	21,052	76,544	1,407,034

- ▶ The mechanism is declared: the category-mapping table was extended about eleven months after the earlier extraction, and the added places fall mostly in Entertainment, Outdoors and Travel.
- ▶ A controlled comparison confirms that **every POI, user and check-in of the earlier extraction reappears in the current one**, which adds others.
- ▶ The three Chapter 3 figures are themselves a declared errata: the published article left placeholders, and the values come from the published CoUrb table.

Cost of the Ch. 3 model

4. B4-5 · 2,3 times the two single-task models, in wall time

Model	Time (s)	Epochs	MFLOPs
Category	16.26	3.8	2.315
Next	18.71	3.2	0.012
MTL	80.88	3.2	0.234

- ▶ Wall time to reach the target F1 scores: **80.88 s against the cumulative 34.97 s** of the two single-task models, about **2.3 times**.
- ▶ MFLOPs do not follow the same pattern, and the chapter says so.
- ▶ **The published prose of that article carries two defective statements about this measurement, both registered in its own errata. I quote the table, not the prose.**

5-fold CV • targets: Category F1 47, Next F1 32.2 • next category (Def. 2.7) • Cap. 3, Tab. 4; erratas em ERRATA.md e Apêndice B do suplemento.

The DGI objective

4. B4-DGI · It may be degenerate, and it was never re-measured

- ▶ An audit recorded, incidentally and outside the leak question, that the contrastive objective **as implemented appears degenerate**: positive and negative score sets identical as multisets, and a measured loss floor matching $2 \cdot \ln 2$ to six decimals.
- ▶ If confirmed, “*trained DGI embedding*” may not describe what Chapter 3 actually used.
- ▶ **What I cannot say is that it was re-measured. It was not.** The artifacts of that audit are not in the repository and the observation was never acted on.
- ▶ **What it would limit:** Chapter 3 is the **null result** of the arc. A weaker representation than assumed makes the null **less** surprising, not more. Neither later chapter inherits DGI: Chapter 4 replaces it, Chapter 5 builds on the check-in-level representation.

ledger §F da auditoria de código, em `material_apx_static_scope.tex`. **não consta em nenhum dos dois volumes.**

Appendix G counts

4. B-APXG · The published 100,2% was counted at the wrong depth

- ▶ **Printed (supplement, Appendix G, Table 8):** joint 4,197,621 (AL) and 5,151,189 (CA); original dedicated 644,359; wider dedicated 4,207,399 and 5,249,719, labeled 100.2% and 101.9% of the joint budget.
- ▶ **Recounted against an independent implementation of the head:** 1,433,863 · 9,634,471 (230%) · 12,044,791 (234%). The published figures counted the same widths at the wrong depth — two layers instead of the four the cell inherits.
- ▶ **The conclusion does not fall. It gets stronger.** The wider arm was not capacity-matched: it received **more than double** the joint budget and still scored **lower** than the narrow model.
- ▶ What does not survive: the labels “100.2% / 101.9% matched”. Table 7’s result columns (Alabama – 0.53, $p = 0.0011$; California, three arms within 0.06) are unaffected.

suplemento, Apêndice G, Tabelas 7 e 8, pp. 24-26; recontagem em NEW_VERSION.md §10.6.
pós-submissão: não consta nos dois volumes.

Clean-room rebuild

4. B-MTLCHECK · Eight cells, mean delta of one thousandth

- ▶ A clean reimplementaion, written without reusing code from the old repository, running from raw check-ins to trained models. **No file produced by the old repository enters the path.**
- ▶ Eight cells at Alabama and Arizona, under **the chapter's own protocol** (five flat folds): **mean delta — 0.001 pp**, largest single deviation **0.421 pp** — **of the order of** the seed spread the chapter itself prints (0.04 to 0.22), about twice its top, and the only cell above it.
- ▶ Two caveats the sentence must carry: **one seed** on the new side against four on the chapter's; and the two columns are **not the same configuration** (representation rebuilt, region tower unified, one component corrected).
- ▶ **This is extra material. It does not correct Chapter 5.**

NEW_VERSION.md §3 e §11; evidência em estado_vs_dissertacao.json. pós-submissão:
não consta nos dois volumes.

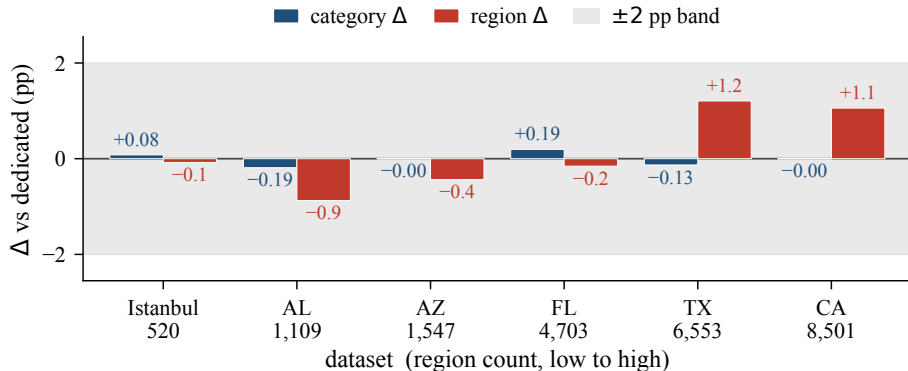
Seed or repetition

4. B-NOM · It names a confound; it does not change the inference

- ▶ **It does not change the inference.** The reported test was always the paired t on the **four per-seed means**: $n = 4$, three degrees of freedom. The registry already separates $n = 20$ (fitted models) from $n = 4$ (inferential unit).
- ▶ **It changes the name, and it names a limit.** Each seed is one **repetition** of the cross-validation: one integer drives **two** things, the user partition and the initialization — and **no result produced so far separates them**.
- ▶ Measured: between folds ≈ 1.2 pp · between repetitions **0.02 to 0.07 pp** · paired band over repetitions **0.05 to 0.15 pp**.
- ▶ The fold term is **20 to 50 times** the repetition term **and is common to both arms** — which is why pairing detects what an unpaired analysis does not.
- ▶ The later work did not correct an error. **It named a confound nobody had named.**

Figure 7

4. B1-6 · The ten differences, with their intervals



figures/mobiwac/fig4_deltas.pdf — Fig. 7, Cap. 5 (volume principal). A figura imprime uma casa decimal na região; a tabela do veredito imprime duas.

Protocol of Ch. 3 and 4

4· B2-2 · Sample-stratified, one repetition, declared in the chapter

- ▶ Declared in Chapter 3 itself (p. 46): a stratified splitter **over the samples**, so one user's check-ins may appear in both training and validation. For the category task the sample unit is the place, so **no place spans two folds**. One pinned seed, so the five folds are **one repetition**.
- ▶ Both prefaces date their conclusions: p. 36, *"Its conclusions are the conclusions of the time, for the configuration studied here"*; p. 52, *"The conclusions reported here are those of the time, for that configuration"*.
- ▶ What each chapter carries forward is **internal and directional**: Chapter 3 delivers a **null result**; Chapter 4 compares against **one baseline under the same protocol on both arms**.
- ▶ No conclusion of this dissertation rests on an absolute number from either.

next category (Def. 2.7) • Chs. 3/4 report **per-category F1**, not macro-F1 • Cap. 3, p. 46-47; prefácios p. 36 e p. 52.

Appendix letters

4. B6-2 · Always name the volume: B is Disclosure here, Errata there

main volume (119 pp)	supplement (27 pp)
A Other Scientific Contributions	not present
B AI-Use Disclosure	Errata to the Reproduced Articles
C Data Ethics and Governance	not present
D Why the Two Tasks Do Not Compete on the Shared Trunk	A Label-History Benchmark for the Next-Category Task
E How Check2HGI and the Joint Model Work	The Human-Subjects Question
F not present	Adaptation of the HGI Baseline
G not present	A Parameter-Count Control for Next-Category Prediction

- The deposited text **does not point at the supplement by design**: it cites only itself and the repository. That is the document's policy.

articles/dissertacao/CLAUDE.md §1; política declarada em
wrapup/erratas/README.md (Q24).

Search coverage

4· B2-5 · Two errata, offered rather than defended

- ▶ **ERR-6.** The clause *“at Florida and California it was not varied, so those two carry the value the smaller searches selected”* credits the small-dataset searches with a value they did not select. The small searches selected **0.0025 and 0.0005**; Florida and California carry **0.005**, the large-dataset tier. The coverage is right; the **provenance clause** is not.
- ▶ **ERR-7.** The same sentence grades **Florida** with Texas as *“fewer folds”* in the batch-size search. In the dedicated category family the single-fold annotation is **Texas only**; Florida’s single-fold screen belongs to the **joint-model** search, described one sentence later.
- ▶ Neither changes a number or a verdict. Both **reduce** what the sentence claims.